English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

English transcription of all twelve handwritten pages. The original calculations and their inconsistencies are preserved; this is a translation, not a corrected solution set.

Problem 1

1. We use Bayes' formula. Drawing a yellow, green, or blue ball means rolling one, two, or three dice, respectively. Thus

$$\begin{aligned}\mathbb{P}(\text{green} \mid \text{sum } 3) &= \frac{\mathbb{P}(\text{sum } 3 \mid \text{green})\mathbb{P}(\text{green})}{\sum_i \mathbb{P}(\text{sum } 3 \mid \text{colour } i)\mathbb{P}(\text{colour } i)} \\ &= \frac{\frac{2}{36}\frac{5}{11}}{\frac{1}{6}\frac{4}{11} + \frac{2}{36}\frac{5}{11} + \frac{1}{6^3}\frac{2}{11}} \approx 29.13\%.\end{aligned}$$

2. The probability that the largest result is at least 3 is

$$\mathbb{P}(\max \geq 3) = 1 - \mathbb{P}(\text{all results} < 3) = 1 - \left(\frac{1}{3}\right)^5 \approx 99.59\%.$$

   For the product to be divisible by 10, let $A = \{\text{product divisible by 2}\}$ and $B = \{\text{product divisible by 5}\}$. Then

$$\begin{aligned}\mathbb{P}(A \cap B) &= 1 - \mathbb{P}(\overline{A} \cup \overline{B}) \\ &= 1 - \mathbb{P}(\overline{A}) - \mathbb{P}(\overline{B}) + \mathbb{P}(\overline{A} \cap \overline{B}) \\ &= 1 - \left(\frac{1}{2}\right)^5 - \left(\frac{5}{6}\right)^5 + \left(\frac{1}{3}\right)^5 \approx 57.10\%.\end{aligned}$$

3. The expected payoff is

$$\begin{aligned}&\left[\left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2\right]\left[\frac{\binom{2}{2}}{\binom{6}{2}}2 + \frac{2\cdot 4}{\binom{6}{2}}1\right] - \frac{4}{9}\left[\frac{1}{2}\cdot 3\right] \\ &\qquad = -\frac{8}{27} < 0.\end{aligned}$$

   For a fair game, replace the payment by x BGN:

$$\frac{10}{27} - \frac{4}{9}\frac{x}{2} = 0 \quad \Longrightarrow \quad \boxed{x = \frac{5}{3}}.$$

Problem 2

1. By the definition of conditional probability,

$$\mathbb{P}(X > m+n \mid X \geq n) = \frac{\mathbb{P}(X > m+n)}{\mathbb{P}(X \geq n)} = \frac{\mathbb{P}(X > m+n)}{\mathbb{P}(X > n-1)}.$$

   For $X \sim \mathrm{Ge}(p)$, with probabilities pq^k, $k \geq 0$,

$$\begin{aligned}\mathbb{P}(X > n) &= \sum_{k=n+1}^{\infty} pq^k = pq^{n+1}(1 + q + \cdots) \\ &= \frac{pq^{n+1}}{1-q} = q^{n+1}.\end{aligned}$$

   Consequently,

$$\frac{q^{m+n+1}}{q^n} = q^{m+1} = \mathbb{P}(X > m).$$

2. The handwritten argument starts with

$$\mathbb{P}(Y > m+n \mid Y > n+1) = \mathbb{P}(Y > m),$$

and then writes, for $n \geq 1$,

$$\mathbb{P}(Y > m+n) = \mathbb{P}(Y > n-1)\mathbb{P}(Y > m).$$

Its recurrence is

$$\begin{aligned}\mathbb{P}(Y > m+1) &= \mathbb{P}(Y > 0)\mathbb{P}(Y > m) \\ &= \mathbb{P}(Y > 0)\big[\mathbb{P}(Y > 0)\mathbb{P}(Y > m-1)\big] \\ &= \cdots = \big(\mathbb{P}(Y > 0)\big)^{m+2}.\end{aligned}$$

Let $\mathbb{P}(Y = 0) = p$. Then $\mathbb{P}(Y > 0) = 1 - p =: q$, and the source writes

$$\begin{aligned}\mathbb{P}(Y = k) &= \mathbb{P}(Y > k-1) - \mathbb{P}(Y > k) \\ &= q^{k+1} - q^k = q^k(1-q) = pq^k.\end{aligned}$$

It concludes that $Y \sim \mathrm{Ge}(\mathbb{P}(Y = 0))$.

3. Using the i.i.d. assumption as in the handwritten solution,

$$\begin{aligned}\mathbb{P}(\min\{X_1, \ldots, X_n\} \leq t) &= 1 - \mathbb{P}(X_{\min} > t) \\ &= 1 - \mathbb{P}(X_1 > t)^n, \\ \mathbb{P}(\max\{X_1, \ldots, X_n\} \leq t) &= \mathbb{P}(\text{all } X_i \leq t) \\ &= \mathbb{P}(X_1 \leq t)^n.\end{aligned}$$

4. Distinguish the five dice. The time until the first six on each die has the geometric distribution with parameter $1/6$. Here use the convention $\mathbb{P}(X = k) = pq^{k-1}$ for $k \geq 1$. Let X_i be the first-six time of die i. Then $K = \max_i X_i$ and

$$\mathbb{P}(K \leq n) = \mathbb{P}(X_1 \leq n)^5.$$

Now

$$\begin{aligned}\mathbb{P}(X \leq n) &= \sum_{k=1}^{n} pq^{k-1} = \sum_{k=0}^{n-1} pq^k = \frac{p(1-q^n)}{1-q} = 1 - q^n, \\ \mathbb{P}(K \leq n) &= (1-q^n)^5, \\ \mathbb{P}(K = n) &= \mathbb{P}(K \leq n) - \mathbb{P}(K \leq n-1) \\ &= \boxed{(1-q^n)^5 - (1-q^{n-1})^5}, \qquad q = \frac{5}{6}.\end{aligned}$$

Problem 3

1. As in the birthday problem, the probability that all n bit vectors of length m are distinct is

$$\boxed{\frac{\prod_{k=0}^{n-1}(2^m - k)}{(2^m)^n}}.$$

The expected total number of ones is

$$\mathbb{E}\,\mathrm{Bin}(nm, \tfrac{1}{2}) = \frac{mn}{2}.$$

The expected number of vectors containing fewer than $m/4$ zeros is $\mathbb{E}\,\mathrm{Bin}(n, p) = np$, where

$$p = \sum_{k=0}^{\lceil m/4 \rceil - 1} \mathbb{P}\big(\mathrm{Bin}(m, \tfrac{1}{2}) = k\big).$$

2. By linearity of expectation, the expected number of equal pairs is

$$\mathbb{E}\sum_{i \neq j} \mathbf{1}_{\{X_i = X_j\}} = \binom{n}{2}\mathbb{P}(X_1 = X_2) = \frac{\binom{n}{2}}{2^m}.$$

For $p \neq 1/2$, the corresponding calculation is

$$\begin{aligned}
&\binom{n}{2}\sum_{k=0}^{m}\binom{m}{k}p^k p^k (1-p)^{m-k}(1-p)^{m-k} \\
&= \binom{n}{2}\sum_{k=0}^{m}\binom{m}{k}(p^2)^k\big((1-p)^2\big)^{m-k} \\
&= \boxed{\binom{n}{2}\big[p^2+(1-p)^2\big]^m}.
\end{aligned}$$

Alternatively, two coordinates agree with probability $(1-p)^2+p^2$. Independence across coordinates gives $\mathbb{P}(X_1 = X_2) = [p^2+(1-p)^2]^m$.

3. The handwritten inclusion–exclusion argument for the expected number of distinct vectors is

$$\begin{aligned}
&\mathbb{E}\big[n - \#\text{equal pairs} + \#\text{equal triples} - \#\text{equal quadruples} + \cdots\big] \\
&= \mathbb{E}\left[\sum_{k=1}^{\infty}(-1)^{k+1}\sum_{\substack{\text{distinct}\\ i_1,\ldots,i_k}}\mathbf{1}_{\{X_{i_1}=\cdots=X_{i_k}\}}\right] \\
&= \sum_{k=1}^{\infty}(-1)^{k+1}\binom{n}{k}\sum_{j=0}^{m}\binom{m}{j}(p^k)^j\big((1-p)^k\big)^{m-j}.
\end{aligned}$$

The source's last line is

$$\sum_{k=1}^{\infty}(-1)^{k+1}\big(p^k+(1-p)^k\big)^m.$$

(The binomial coefficient in the preceding line is absent from that last handwritten line.)

4. We seek the least n for which the probability that all vectors are distinct is less than $1/2$.

Method I.

$$\begin{aligned}
&\prod_{k=0}^{n-1}\left(1-\frac{k}{2^m}\right) < \frac{1}{2} \\
\iff\quad &\log\prod_{k=0}^{n-1}\left(1-\frac{k}{2^m}\right) < -\log 2 \\
\iff\quad &\sum_{k=0}^{n-1}\log\left(1-\frac{k}{2^m}\right) \approx -\frac{n(n-1)}{2^{m+1}} < -\log 2.
\end{aligned}$$

Here $\log(1+x) \approx x$ for small x. Thus

$$n(n-1) > 2^{m+1}\log 2, \qquad n_{\min} \approx \frac{1+\sqrt{1+2^{m+3}\log 2}}{2} \approx \sqrt{2^{m+1}\log 2}.$$

Method II.

$$\frac{(2^m)!}{(2^m-n)!}\frac{1}{2^{mn}} < \frac{1}{2}.$$

Use Stirling's approximation $n! \approx \sqrt{2\pi n}(n/e)^n$:

$$\begin{aligned}
\frac{(2^m)!}{(2^m-n)!}\frac{1}{2^{mn}} &\approx \frac{\sqrt{2\pi 2^m}(2^m/e)^{2^m}}{\sqrt{2\pi(2^m-n)}((2^m-n)/e)^{2^m-n}}\frac{1}{2^{mn}} \\
&= \sqrt{\frac{1}{1-n/2^m}}\left(\frac{2^m}{2^m-n}\right)^{2^m-n}2^{mn}\frac{1}{2^{mn}}\frac{1}{e^n}.
\end{aligned}$$

For $n \ll 2^m$, the source continues

$$\approx \left(\frac{1}{1-n/2^m}\right)^{2^m-n}\frac{1}{e^n}$$

$$= \frac{(1-n/2^m)^n}{(1-n/2^m)^{2^m}} \frac{1}{e^n} \approx (1-n/2^m)^n < \frac{1}{2}.$$

Therefore

$$\log(1-n/2^m) < -\frac{\log 2}{n} \quad \Longrightarrow \quad \frac{n}{2^m} > \frac{\log 2}{n} \quad \Longrightarrow \quad n > \sqrt{2^m \log 2}.$$

Where did a $\sqrt{2}$ go? In both cases, $n_{\min} = \Theta(2^{m/2})$ as $m \to \infty$.

Problem 4

The natural strategy is to choose *higher* for 1 and 2 and *lower* for 4 and 5; for 3 either choice is equivalent. For simplicity, take *lower* for 3.

Since the probabilities are uniform, we can simply count the combinations leading to each winning state. The original table is reproduced below. The final column was left empty in the handwritten notes.

number m\move n	1	2	3	4	5
1	1	3	9	28	
2	1	4	12	37	
3	1	4	12	37	
4	1	3	9	28	
5	1	2	7	21	
probability of reaching move n	$\frac{5}{5}$	$\frac{16}{5\cdot4}$	$\frac{49}{5\cdot4\cdot4}$	$\frac{151}{5\cdot4\cdot4\cdot4}$	

The coloured arrows in the original show how the counts in the next column are obtained by adding counts from the admissible preceding states.

If we play until move k, the expected payoff is the probability of reaching that move multiplied by the payoff at move k. See the notebook for a generalization.

If stopping at any time is also allowed (which is the realistic scenario), the optimal expected payoff at each state is

$$\max \left\{ \text{current payoff}, \sum_{i,k} \mathbb{P}(\text{move to state } (i,k)) \, \text{optimal payoff at } (i,k) \right\}.$$

The partially filled table in the handwritten solution is

A	1	2	3	4
1			1000	1000
2			750	1000
3				1000
4				1000
5				1000
loss				

For example,

$$A[1,3] = \max \left\{ 100, \frac{1}{4}\,1000 + \frac{1}{4}\,1000 + \cdots \right\} = 1000,$$
$$A[2,3] = \max \left\{ 100, \frac{1}{4}\,0 + \frac{3}{4}\,1000 \right\} = 750.$$

The first entry inside the maximum is the payoff if we stop; the second is the expected payoff if we continue. In the second example, rolling a 1 means losing. Continue in the same way. See the notebook for the solution of the general case.